Venkata Lakshmi Parimala Pasupuleti

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github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

AI Engineer with 9 years of production Python engineering and hands-on enterprise GenAI development. Built **AgentProbe** -- an open-source agentic AI toolkit with a 3-agent LangGraph runtime, ChromaDB vector search pipeline, and 5-metric LLM-as-judge evaluation loops covering factual accuracy, hallucination, retrieval, completeness, and compliance. Delivered **AI on FHIR** -- a HIPAA-compliant NLP platform for healthcare structured medical records. Applied responsible AI practices across regulated financial and healthcare data at scale (1.6M+ records, 100K+ MAU). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

George Mason University
Aug 2024 -- Aug 2025

Research Associate -- Virginia, USA

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries -- multi-source data collection, transformation, and stakeholder reporting.
- Applied Isolation Forest to automate data-quality pipelines across 3 research projects -- surfaced anomalies manual review missed; designed Tableau and ggplot2 dashboards improving research comprehension by 40%.

Entrata India (Db Xento Systems)
Jan 2017 -- Aug 2024

Software Engineer -- Pune, India

- Applied Python, PyTorch, and OpenAI API to build scalable GenAI data transformation pipelines across **1.6M financial records** -- production AI on regulated data with output validation, quality gates, and AI governance controls for enterprise financial workflows.
- Reduced deployment times by 40% by building 20+ full-stack microservices with enterprise API integration (Python, PostgreSQL, RabbitMQ, FastAPI) for 100K+ MAU across 120+ Agile sprints; cut execution time 50% via Docker containerization; improved query performance 30% across 200+ SQL queries.
- Achieved 100% PHI compliance migrating all client data to AWS S3 with Libsodium-encrypted pipelines -- zero security incidents; led 300+ peer code reviews mentoring engineers on responsible AI and production best practices.

PROJECTS

AgentProbe: Agentic AI Testing Toolkit
In Progress

Python · LangChain · LangGraph · LangSmith · FastAPI · ChromaDB · Anthropic · OpenAI · Groq · React · Docker · GitHub Actions -- github.com/venkatalakshmiparimala/AgentProbe

- Built a 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) implementing agent-based workflows for task orchestration -- tool calling, inter-agent state handoffs, multi-provider model routing (Anthropic, OpenAI, Groq, Ollama) deployed as a 15-endpoint FastAPI backend using open-source frameworks at zero cost.
- Automated quality gates with a **5-metric LLM-as-judge evaluation loop** (factual accuracy, hallucination, semantic retrieval, completeness, compliance) with CI/CD integration -- failing agent runs blocked before production; end-to-end observability via React dashboard tracking vector search quality, latency, and regression drift.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · spaCy · BERT · Embeddings · Docker -- github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced clinical retrieval time by ~60% by building an NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and semantic retrieval -- **HIPAA-compliant RBAC** with AI governance controls across structured healthcare medical records.

PropLens: Lease Intelligence & Compliance AI
In Progress

Python · LangGraph · ChromaDB · FastAPI · LLMs · React

- Building agent-based RAG pipeline over lease PDFs -- vector search and semantic retrieval across multi-unit portfolios; LLM-as-judge scoring with hyperparameter-tuned anomaly detection across 1,000+ tenant records.

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — 2024 -- 2026

GPA: 3.87 · Fairfax, Virginia

B.Tech Computer Science & Engineering -- Sree Venkateshwara College of Engineering, India — 2013 -- 2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, LangSmith, Multi-Agent Orchestration, Agent-Based Workflows, Task Orchestration, ChromaDB, RAG Pipelines, Vector Search, Semantic Retrieval, Embeddings, LLM Evaluation, LLM-as-Judge, Evaluation Loops, Eval Frameworks, Agent Observability, Prompt Engineering, AI Governance, Responsible AI, OpenAI API, Anthropic API, Groq, Hugging Face Ecosystem, FastAPI

ML & Data: PyTorch, TensorFlow, XGBoost, Scikit-learn, Isolation Forest, Random Forest, Hyperparameter Tuning, NumPy, Pandas, spaCy, BERT, Tableau, Power BI, ggplot2

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ, PostgreSQL, MySQL, Oracle